

SMART ATTENDANCE SYSTEM

- Using ESP32-CAM, Face Recognition & Fingerprint Authentication
- B.E. Mechatronics Engineering | The Oxford College of Engineering
- Yogesh R • Yathish • Jojin Alex • Mohammed Shadab • Sachin K

1. Introduction

- Automated attendance system designed to improve security, accuracy and efficiency.
- Combines computer vision, embedded systems, IoT and biometric authentication.
- Uses face recognition and fingerprint verification to reduce proxy attendance and manual errors.
- Attendance is recorded digitally with date and time.

2. Objectives

- Automatically detect and identify registered users through live video.
- Provide fast biometric attendance using face recognition and fingerprint verification.
- Prevent unauthorized attendance through dual verification.
- Store attendance records automatically with date and time.
- Provide real-time status feedback through an OLED display.

3. Problem & Motivation

- Traditional attendance is time-consuming and prone to manual errors.
- Proxy attendance and false entries reduce reliability.
- A second biometric layer strengthens identity verification.
- The project demonstrates a practical, low-cost smart automation solution.

4. Hardware Components

- ESP32-CAM — captures live video and provides Wi-Fi connectivity.
- ESP32 DevKit V1 — main controller for fingerprint, OLED and wireless coordination.
- R307 Fingerprint Sensor — stores and verifies fingerprint templates.
- SSD1306 OLED Display — shows user and authentication status in real time.
- Stable 5V supply with common ground and UART/I²C communication.

5. Software Components

- Python application for face recognition and attendance processing.
- OpenCV for image and video processing.
- Face Recognition library for facial encoding and comparison.
- NumPy and Requests for supporting image/data operations.
- Arduino IDE for ESP32 firmware development.
- CSV database for attendance records.

6. System Architecture

- ESP32-CAM captures the student's face and communicates over Wi-Fi.
- Python receives the video/image stream and compares the face with registered encodings.
- ESP32 DevKit communicates with the R307 sensor through UART.
- OLED provides real-time feedback such as Place Finger, Scanning Face, Attendance Marked and Access Denied.
- Successful biometric verification results in automatic attendance logging.

7. Project Flow

1. ESP32-CAM starts and provides the live video stream.
2. Python detects a face and identifies the registered student.
3. The identified Face ID is retained for verification.
4. The system requests fingerprint verification through the ESP32/R307.
5. Fingerprint ID is returned and compared with the Face ID.
6. Only when both IDs match is attendance marked.

8. Dual Authentication Logic

- Face ID = Fingerprint ID → Attendance MARKED.
- Face recognized but fingerprint does not match → Attendance NOT MARKED.
- Unknown face → Authentication fails.
- Unknown fingerprint → Authentication fails.
- This two-factor approach helps prevent proxy attendance and unauthorized entries.

9. Attendance Recording

- After successful face + fingerprint verification, the student's attendance is recorded.
- The record contains the student's identity along with date and time.
- The report describes CSV-based digital storage for attendance records.
- The system removes the need for manual attendance entry.

10. Outcomes

- Dual biometric authentication implemented using fingerprint and face recognition.
- Reduced risk of proxy attendance and unauthorized access.
- Wireless communication between ESP32 DevKit V1 and ESP32-CAM simplifies wiring.
- Automated attendance logging reduces manual work.
- OLED provides immediate user feedback.
- Prototype uses affordable and scalable components.

11. Advantages & Applications

- Advantages: automatic marking, dual verification, real-time feedback, low cost and easy upgradeability.
- Schools and colleges — student attendance management.
- Offices and companies — employee attendance tracking.
- Industries and factories — worker attendance and access control.
- Laboratories and secure facilities — authorized-user verification.

12. Future Scope

- Cloud integration for remote attendance monitoring.
- Google Cloud/Firebase or other online databases for centralized storage.
- Android/iOS application for administrators and notifications.
- AI-based recognition improvements for different angles and lighting.
- Web dashboard with graphical attendance analytics.
- Additional RFID/NFC authentication and stronger security.

13. Conclusion

- The project integrates face recognition and fingerprint authentication into one attendance platform.
- Dual-layer verification improves reliability and reduces proxy attendance.
- ESP32-CAM, ESP32 DevKit, R307 and OLED work together with Python-based computer vision.
- The system demonstrates practical integration of AI, IoT, embedded systems and biometric security.
- The design can be extended toward cloud-connected, large-scale attendance management.

14. References

- Turk & Pentland — Face Recognition Using Eigenfaces (1991).
- Viola & Jones — Rapid Object Detection Using a Boosted Cascade of Simple Features (2001).
- Zhao et al. — Face Recognition: A Literature Survey (2003).
- Jain, Ross & Prabhakar — An Introduction to Biometric Recognition (2004).
- Parkhi, Vedaldi & Zisserman — Deep Face Recognition (2015).
- Kortli et al. — Face Recognition Systems: A Survey (2020).
- OpenCV, ESP32-CAM, R307 and Arduino documentation.

THANK YOU

Questions & Discussion